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JAVA PROJECT REPORT ON

Warehouse Inventory Monitoring and Stock Reorder System

Submitted by

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DECLARATION

I hereby declare that the project entitled " Warehouse Inventory Monitoring System" submitted in partial fulfillment of the requirements for the course JAVA PROGRAMMING is a record of my original work carried out under the guidance of , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

MADHUSHREE K (Your Name)

CERTIFICATE

This is to certify that the project entitled Warehouse Inventory Monitoring and Stock Reorder System" is a bona fide work carried out by [MADHUSHREE K] [I-BSc IT-'A'], [], KG College of Arts & Science, in partial fulfillment of the requirements for the course [JAVA PROGRAMMING].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place: COIMBATORE

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Submitted for the Viva-Voce Examination held on KG COLLAGE OF ARTS & SCIENCE

Guide: Head of the Department

(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

TheWarehouse Inventory Monitoring and Stock Reorder Systemis a Java-based application designed to automate the process of managing inventory in warehouses. In many small businesses and storage facilities, inventory operations are carried out manually, which often leads to errors in stock calculation, delays in tracking products, and inefficient record management. This project aims to simplify inventory handling by providing an automated system that monitors stock levels and manages product details effectively.

The system allows users to add products, view available inventory, update stock quantities, and automatically identify items that fall below the reorder level. It ensures that low-stock products are detected in time, helping in proper stock replenishment and avoiding shortages. The application is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and basic data structures like ArrayList.

The main objective of the system is to improve inventory management efficiency, reduce human errors, and provide accurate and real-time stock information. The system generates a clear view of inventory details including product ID, name, quantity, and reorder level, enabling better decision-making.

This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a strong foundation for developing more advanced inventory and warehouse management systems in the future.

1. INTRODUCTION

Warehouse inventory management is an essential part of business operations.

It helps in tracking goods, maintaining stock levels, and avoiding shortages.

Traditional methods of inventory management involve manual record keeping.

These methods are time-consuming and prone to human errors.

Automation improves efficiency and accuracy in managing stock.

This project provides a console-based solution using Java.

It allows easy monitoring and updating of inventory details.

The system ensures proper stock control and timely reordering.

1.1 Objective

- The main objective is to develop a simple inventory system.

  It aims to manage warehouse products efficiently.

  The system helps track stock levels in real-time.

  It reduces errors caused by manual calculations.

  It provides alerts when stock reaches minimum levels.

  It ensures timely restocking of products.

  It demonstrates Java programming concepts practically.

  It improves overall inventory management efficiency.

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1.2 Overview

This system is a console-based Java application.

It uses a menu-driven interface for user interaction.

Users can add, update, and view product details.

The system stores product information in memory.

It uses ArrayList for efficient data handling.

Stock levels are monitored continuously.

Low stock items are identified automatically.

The system is simple, flexible, and easy to use.

1.3 Features

The system allows adding new products easily.

It displays all available inventory records.

It supports updating stock quantities.

It provides alerts for low stock levels.

The interface is simple and user-friendly.

It uses Java collections for data storage.

It performs operations quickly and efficiently.

It reduces manual workload significantly.

2. PROGRAM ANALYSIS

2.1 Problem Definition

Manual inventory systems are inefficient and slow.

They require a lot of human effort and time.

Errors in calculation may lead to losses.

Tracking stock manually is difficult and unreliable.

There is no automatic alert for low stock.

Data management becomes complex over time.

Businesses need a better solution for inventory control.

Hence, an automated system is required.

2.2 Project Overview

The project is developed using Java programming language.

It follows object-oriented programming concepts.

The system manages products using classes and objects.

ArrayList is used to store product data.

Scanner is used to accept user input.

It includes modules for different operations.

The system is menu-driven for easy usage.

It ensures smooth and efficient inventory management.

2.3 Scope and Limitations

The system is useful for small-scale warehouses.

It helps in managing inventory effectively.

It improves accuracy in stock tracking.

It reduces manual work and effort.

However, it does not store data permanently.

The system does not include a graphical interface.

It supports only basic operations.

Future improvements can enhance its capabilities.

3. SYSTEM DESIGN

3.1 Product Module

The product module is responsible for handling product details.

Each product has attributes like ID, name, quantity, and reorder level.

It uses a class structure to represent product objects.

The module ensures that each product is uniquely identified.

It stores product information in an organized format.

It supports adding new products into the system.

The module helps in easy retrieval of product details.

It forms the foundation of the entire inventory system.

3.2 Inventory Module

The inventory module manages all product records.

It uses ArrayList to store multiple products dynamically.

It allows efficient storage and retrieval of data.

The module maintains the current stock details.

It helps in displaying the complete inventory list.

It ensures data consistency during operations.

It supports adding and updating product information.

It plays a key role in overall system functionality.

3.3 Stock Update Module

The stock update module is used to modify product quantity.

It allows users to update stock based on new data.

The module searches for products using product ID.

It ensures that only valid products are updated.

It updates the quantity field of the selected product.

It provides confirmation after successful update.

It prevents incorrect data entry.

It ensures accurate and up-to-date inventory records.

3.4 Stock Monitoring Module

The stock monitoring module checks inventory levels.

It compares product quantity with reorder level.

It identifies products that are low in stock.

It generates alerts for such products.

It helps in preventing stock shortages.

It supports better inventory planning.

It ensures timely reordering of products.

It improves overall stock management efficiency.

3.5 User Interface Module

The user interface module handles user interaction.

It provides a menu-driven console interface.

It allows users to select different operations.

It displays input prompts and output results.

It ensures smooth communication between user and system.

It is simple and easy to understand.

It guides users through each step clearly.

It enhances usability of the application.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Warehouse Inventory Monitoring System , a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

The DDRMS follows a structured development approach.

4.1 Problem Analysis

The problem involves managing inventory efficiently.

Manual systems create delays and errors.

There is a need for automation in stock handling.

The system should track and update inventory.

It should alert when stock is low.

The solution must be simple and effective.

Java is chosen due to its flexibility.

The system is designed based on these needs.

4.2 System Design

The system uses object-oriented design principles.

It is divided into different functional modules.

Each module performs a specific task.

The design ensures easy understanding and usage.

Menu-driven structure simplifies navigation.

Data is stored using ArrayList.

The system is designed for scalability.

It ensures smooth execution of operations.

4.3 Program Development

The program is developed using Java language.

Basic programming concepts are used effectively.

Classes and objects are used for structure.

Loops and conditions control the program flow.

Scanner is used for user input.

ArrayList stores product data dynamically.

The code is simple and easy to understand.

Proper testing is done during development.

4.4 Testing and Validation

Each module is tested individually.

Inputs are verified for correctness.

Outputs are checked for accuracy.

Stock updates are validated properly.

Low stock alerts are tested thoroughly.

Errors are identified and corrected.

The system performs all operations correctly.

Final testing ensures system reliability.

4.5 Result and Documentation

The system works as expected successfully.

It stores and manages product data efficiently.

Stock levels are updated correctly.

Low stock items are identified easily.

The results are accurate and reliable.

Documentation explains system functionality clearly.

All modules are properly described.

The project meets its objectives effectively.

4.6 Algorithm

Start the program execution.

Display the menu options to the user.

Accept the user’s choice as input.

Perform the selected operation.

Add, view, or update product details.

Check for low stock conditions.

Repeat the process until exit.

Stop the program when user exits.

4.7 Flowchart

The flowchart begins with the start symbol.

It displays menu options to the user.

User selects an operation.

Decision-making process is applied.

Operations like add, view, update are executed.

Stock checking is performed if selected.

The process repeats continuously.

Finally, the program ends when exit is chosen.

5. SOURCE CODE

```
import java.util.*;

public class WarehouseSystem {
   static Scanner sc = new Scanner(System.in);
   static List<int[]> inv = new ArrayList<>(); // {id, qty, reorder}, name in separate list
   static List<String> names = new ArrayList<>();

   public static void main(String[] args) {
      while(true){
         System.out.println("\n1.Add 2.View 3.Update 4.Low 5.Exit");
         switch(sc.nextInt()){
            case 1: add(); break;
            case 2: view(); break;
            case 3: update(); break;
            case 4: low(); break;
```

```
            case 5: System.exit(0);
          }
        }
    }

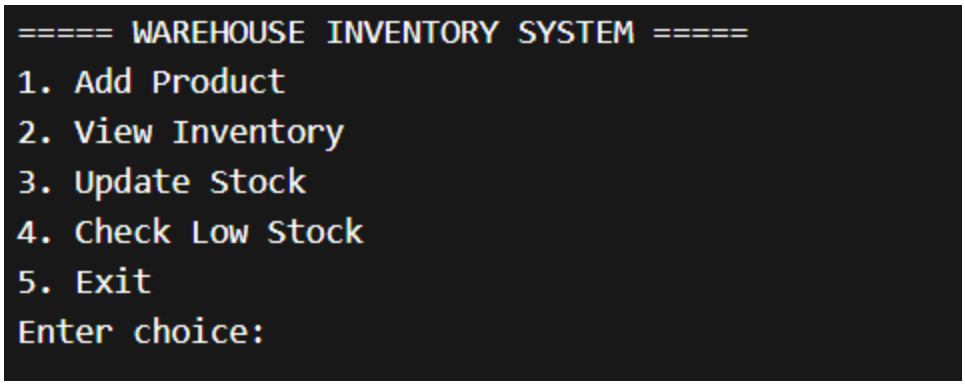
    static void add(){
        System.out.print("ID: "); int id=sc.nextInt(); sc.nextLine();
        System.out.print("Name: "); String n=sc.nextLine();
        System.out.print("Qty: "); int q=sc.nextInt();
        System.out.print("Reorder: "); int r=sc.nextInt();
        inv.add(new int[]{id,q,r}); names.add(n);
    }

    static void view(){
        for(int i=0;i<inv.size();i++)
            System.out.println(inv.get(i)[0]+" "+names.get(i)+" Qty:"+inv.get(i)[1]+" Reorder:"+inv.get(i)
[2]);
    }

    static void update(){
        System.out.print("ID: "); int id=sc.nextInt();
        for(int[] p:inv) if(p[0]==id){ System.out.print("New Qty: "); p[1]=sc.nextInt(); return; }
        System.out.println("Not found");
    }

    static void low(){
        for(int i=0;i<inv.size();i++) if(inv.get(i)[1]<=inv.get(i)[2])
            System.out.println(names.get(i)+" low! Qty:"+inv.get(i)[1]);
    }
}
```

6. INPUT AND OUTPUT SCREENS



```
Enter choice: 1

Enter Product ID: 101
Enter Product Name: Rice
Enter Quantity: 50
Enter Reorder Level: 20

Product added successfully!
```

```
Enter choice: 2

--- Inventory List ---
ID: 101, Name: Rice, Quantity: 50, Reorder Level: 20
ID: 102, Name: Sugar, Quantity: 30, Reorder Level: 15
```

```
Enter choice: 3

Enter Product ID to update: 101
Enter new quantity: 80

Stock updated successfully!
```

```
Enter choice: 4

--- Low Stock Products ---
Product: Sugar | Quantity: 10 (Reorder Needed!)
```

```
Enter choice: 4

--- Low Stock Products ---
All products are sufficiently stocked.
```

```
Enter choice: 3

Enter Product ID to update: 999

Product not found!
```

```
Enter choice: 5

Exiting system...
```

7. DISCUSSION & CONCLUSION

The project successfully meets its objectives.

It simplifies warehouse inventory management.

It ensures accurate stock tracking.

It provides timely reorder alerts.

The system improves efficiency and productivity.

It demonstrates practical Java implementation.

The project is useful for learning purposes.

It can be enhanced for real-world applications.

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8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – Java: The Complete Reference

Websites

• https://www.oracle.com/java

• https://www.geeksforgeeks.org

• https://www.w3schools.com

• https://www.javatpoint.com